PHILIP NIRON NITHIANANDAN

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PROFILE

Software Engineer with 4 years of production backend experience across enterprise and aerospace software development. Skilled in Java backend development, AWS cloud infrastructure, and applied AI engineering. Recently completed a Master of Software Engineering in New Zealand, with a peer-reviewed IEEE MIPR 2026 research contribution combining vision transformers, RAG, and large language models.

SKILLS

Programming: Java, Python, JavaScript
Backend: Spring Boot, Java EE, Flask
Frontend: JavaScript, HTML5/CSS3, JSP
Database: PL/SQL, MySQL
Cloud & DevOps: AWS, GCP, Docker, Git, CI/CD
Test Automation: Selenium, Cucumber, Mockito, REST Assured, JUnit
AI/ML: PyTorch, Vision Transformers, Contrastive Learning, RAG, LLM Integration, Prompt Engineering
Data Analytics: Power BI
AI Tools: Amazon Bedrock, Claude Code, GitHub Copilot, Cursor

WORK EXPERIENCE

SOFTWARE ENGINEER - R&D - AEROSPACE & DEFENCE
IFS R&D International (pvt) Ltd
Colombo, Sri Lanka
October, 2023 - July, 2025

- Delivered end-to-end feature development across the full software development lifecycle for enterprise applications in the aerospace and defence domain.
- Built and maintained backend solutions using Java EE, PL/SQL, Google Guice, Gradle, and JMS Queue within an aerospace and defence asset management system (Maintenix).
- Recognised by management for completing assigned JIRA tickets with zero functional bugs reported throughout the tenure.
- Developed automated tests using Selenium, Cucumber, JUnit, Mockito, and REST Assured.
- Collaborated with an international R&D team across multiple time zones to deliver complex enterprise features.

ASSOCIATE ENGINEER - TECHNOLOGY (JAVA)
Virtusa (pvt) Ltd
Colombo, Sri Lanka
October, 2021 - September, 2023

- Analysed software designs and requirements, and developed features in the application.
- Utilized a diverse range of technologies such as Spring Boot, GraphQL, and MySQL.
- Performed defect analysis and implemented effective solutions.
- Interacted directly with the offshore client for requirement clarifications.

EDUCATION

Master of Software Engineering

- Yoobee College of Creative Innovation, New Zealand
- July, 2025 - July, 2026
- Research Area & Capstone Project - ***RadLens: Radiology Report Generation Using a Vision Transformer and Retrieval-Augmented Generation.***

BSc (Hons) Computer Science and Software Engineering

- Grade - First class honours
- University of Bedfordshire, United Kingdom
- January, 2021 - October, 2021
- Research Area & Undergraduate Project - ***CaptureCorn: Android Application to detect Fall Army worm attack in Corn leaves using Convolutional neural network (CNN).***

Higher Diploma in Information Technology

- SLIIT Academy, Sri Lanka
- January, 2018 - December, 2020

POSTGRADUATE ACADEMIC PROJECTS

RadLens: AI-Assisted Radiology Report Generation (2026)

Skills - Python, Flask, PyTorch, AWS, Docker, Vision Transformers, RAG, Prompt Engineering, Amazon Bedrock, Hugging Face Transformers, Kaggle

- Designed and built an AI pipeline that generates structured draft radiology reports from chest X-ray images using vision transformer models and Claude Sonnet through Amazon Bedrock.
- Trained and evaluated BEiT, Swin Transformer, and ViT models on the Indiana University Chest X-Ray dataset, implementing a RAG-based retrieval approach using fine-tuned BEiT visual features that improved BLEU-1 scores by 82% to 124% and achieved a BERTScore F1 of 0.8701.
- Built a Flask-based radiologist review system with inline editing, audit trail, approval workflow, and PDF export.
- Deployed the system on AWS using ECS Fargate, API Gateway, S3, IAM, and CloudWatch.
- Research accepted at the IEEE MIPR 2026 Health-MM Workshop in Bangkok, August 2026.

Care N Cure: AI-Powered Medical Analysis (2025)

Skills - Python, Flask, React, MySQL, GCP, RAG, GEMINI LLM Integration

- Developed a full-stack AI-powered triage and patient management web application for emergency unit workflows as a postgraduate coursework project, simulating real hospital emergency department operations.
- Integrated Google Gemini LLM with a Retrieval-Augmented Generation pipeline using a medical knowledge base to analyse patient symptoms and vital signs, generating severity classifications and specialist referral recommendations.
- Built role-based dashboards for emergency unit receptionists and doctors with real-time patient status updates using React and Flask.
- Deployed on Google Cloud Platform with MySQL database backend.

REFERENCES AVAILABLE UPON REQUEST